

Materials in the DARJa repository

Data and data preparation

The R file darja_data4.RData, which contains an object called `d`, has information on geographical IDs, geographical coordinates, feature type, feature, and feature values. A corresponding tab-delimited text file with the same rows is offered solely for those who do not wish to engage with R; it's called darja_data4.txt, and is given as a zip file.

Files used in the preparation of darja_data4.RData are: DARJa_data_preparation3.R, data-small-final2.csv, feature_values_changes.txt, Cintaksis.xlsx, Fonetika.xlsx, Leksika.xlsx, Morfologiya.xlsx. The last four were prepared by a team at Kazan Federal University and downloaded by us on 2022-03-30 (they are still available at <https://kpfu.ru/atributivnaya-baza-dannyh-russkih-govorov-269324.html> as of 2025-07-21). The Kazan files have IDs for locations, coordinates in an unidentified system, and features and their values. Using the script DARJa_data_preparation3.R we extracted this information and combined it with information on geographical coordinates gathered by ourselves through georeferencing the generic DARJa map—information contained in data-small-final2.csv. We moreover corrected errors relating to names of features and values in the Excel files through a guide to corrections contained in feature_values_changes.txt. None of the files DARJa_data_preparation3.R, data-small-final2.csv, feature_values_changes.txt, Cintaksis.xlsx, Fonetika.xlsx, Leksika.xlsx, Morfologiya.xlsx are used in subsequent analyses, but they are supplied in order to document the nature of the original data and details of the data extraction. Note that locations with the IDs 628 and 2329 do not have proper numbers on the DARJa maps, and their map numbers are given as 0 or -2 in the Kazan files, even if the files do have features and feature values associated with them in those files. They are considered spurious and were deleted from our data, which then has 4193 rather than 4195 locations.

Computing linguistic distances

The R file linguistic_distance_matrix5.RData, which contains an R object called `m`, contains linguistic distances among the 4193 locations. Row and column names correspond to the IDs, so they are sequences from 1 to 4195 with 628 and 2329 missing. The distances were computed using linguistic_distances_darja3.R. That script also contains a toy example for demonstration of how the calculation works. Because of constraints on the size of files at GitHub, access to the linguistic distance matrix is provided as a link in the file distance_matrices_links.txt.

Computing geographical distances

The R file geographical_distance_matrix3.RData, which contains an R object called `mgeo`, contains geographical distances among the 4193 locations. Row and column names correspond to the IDs, so they are sequences from 1 to 4195 with 628 and 2329 missing. The

distances were computed using `geographical_distance_matrix.R`. Because of constraints on the size of files at GitHub, access is provided as a link in the file `distance_matrices_links.txt`.

Clusterization and cluster validation

We produced ultrametric trees based on the distance matrix, using all methods available in the `hclust()` function of base R, except ‘single’, ‘WPGMC’, and ‘UPGMC’, which put nearly all locations in one big cluster even when $k = 5$, for instance. The methods used were: ‘ward.D’, ‘ward.D2’, ‘complete’, ‘average’ (same as UPGMA), and ‘mcquitty’ (same as WPGMA). The trees are in the files `WardD.RData`, `WardD2.RData`, `complete.RData`, `UPGMA.RData`, `WPGMA.RData`, and the objects contained in the files have the same names as the files minus the suffix (so: `WardD`, `WardD2`, etc.). Clusterization was carried out by `clusterization_darja2.R`.

Experiments with cluster validation were directed at finding some quantitative criteria for deciding between the 5 methods and some optimal number of clusters. All experiments are carried out using `optimal_number_clusters5.R`. The first section of the script implements using the Adjusted Rand index (ARI) to quantify the fit between the distribution of feature values over locations (dialects) and a given clusterization. Output of this is in `optimal_number_clusters3.txt`. The second section does silhouette scores on the clusterizations. Output is in `optimal_number_clusters_sil2.txt`. The third section does stability scores, output of which is in `stability_scores2.txt`.

Maps

The `dmap()` function of the `dmap.R` script serves to produce different kinds of maps, some coming with an accompanying text file: custom maps of locations clustered according to different methods and different numbers of clusters, maps indicating geographical cluster outliers, a generic map showing only locations, a map of biomes, and custom maps of the distribution of individual clusters or cluster values.

It requires the files `darja_data4.RData`, `linguistic_distance_matrix5.RData`, `WardD.RData`, `WardD2.RData`, `WardD2`, `complete.RData`, `UPGMA.RData`, `WPGMA.RData`, `cities.RData`, `biomes_locs.txt`, and `locations.R`. A separate file with instructions is provided as `dmap_manual.pdf`.

Biomes

The script `biomes2.R` reads biome data from Olson & Dinerstein (1998) posted at <https://www.worldwildlife.org/publications/terrestrial-ecoregions-of-the-world#:~:text=There%20are%20867%20terrestrial%20ecoregions,conserve%20biodiversity%20around%20the%20world>. The files are in https://files.worldwildlife.org/wwfcmprod/files/Publication/file/6kechn7e3u_official_teow.zip. Unzipped, this will yield a folder, which should be renamed to `official_teow` and be placed as a subfolder of the folder where `biomes.R` is found. The script `biomes2.R` then identifies the biome associated with each

location in DARJa and creates a data frame with IDs, coordinates, and a number corresponding to the biome of each location. This is output to the file `biomes_locs.txt`. The numbers are associated with descriptions in the little file called `key_biomes98.txt`. Subsequently, `biomes2.R` creates a tab-delimited file called `biomes4.txt` with a row per pair of locations giving the location IDs, the geographical and linguistic distances, and a 0 for 'same' and 1 for 'different' biome. This large file is shared via a link in the `biome_data_link.txt`; its contents is further analyzed, also using `biomes2.R`. This script investigates the effect of biomes on linguistic distance when controlling for geographical distance. Results are in `biome_results.txt` and Fig11 `ling_geo_dist_biomes.tiff`.

Rivers

Using `rivers3.R`, the file `rivers_1_all2.txt` was produced. The script and the text file (the latter zipped, and given as a link in `rivers_data_link.txt`) are provided. The file lists all pairs of locations with their linguistic and geographical distances, indicating the number of rivers or lakes that separate them. The river data come from <https://www.naturalearthdata.com/downloads/> via the interface of the `rnaturalearth` R package. The '1' in the file refers to the large-scale data of `naturalearth`. The resolution is 1:10 mill. The data in `rivers_1_all2.txt` are read and analyzed using `analyze_river_data3.R`.

Administrative divisions

All relevant code is in `administrative_units2.R`. This creates a file called `admin_units.txt2`, which has a row per pair of locations giving the location IDs, the geographical and linguistic distances, and a 0 for 'same' and 1 for 'different' oblast as well as a 0 for 'same' and 1 for 'different' rajon. This large file is shared via a link in `admin_divisions_data_link.txt`.

All predictors combined

The code for the combined model of all predictors and the barplot of their relative contributions is in `combined_model2.R`.